

VirtuBox Data Analyst Assessment

Online Retail II — Management Analysis

UCI Machine Learning Repository — <https://archive.ics.uci.edu/dataset/502/online+retail+ii>

Executive Summary

The analysis uses 1,067,371 raw Online Retail II records and an analysis-ready dataset of 1,027,017 rows after documented cleaning. Total analyzed sales are £19,014,209.84. Key findings concern UK concentration, November seasonality, high-value products/customers and returns.

Methodology

Combined the two yearly worksheets; standardized identifiers and text; handled missing Description and CustomerID; removed 34,335 exact duplicates; removed 6,019 non-positive-price rows; created SalesAmount = Quantity × Price; retained negative quantities for operational analysis.

Key Findings

UK generated £16,144,736.48 (84.91%). November was the strongest recurring month. StockCode 22423 generated £314,045.02. The top three identified customers generated £1,389,786.12. Negative-quantity rows were about 1.86% of final records.

Surprising Result

December 2011 appears weak at £432,719.06 versus £1,456,145.80 in November, but the dataset ends on 9 December 2011. Therefore December 2011 is incomplete and cannot be used as a full-month seasonal comparison.

Recommendations

Prepare inventory and campaigns before November; investigate returns/cancellations by product, market and month; develop selected international markets while protecting UK performance.

Limitations

243,007 records have missing CustomerID. December 2011 is incomplete. The dataset has no product cost, gross margin, CAC or fulfillment cost, so profitability cannot safely be inferred.

AI Use

ChatGPT was used for assessment structuring, analytical-question brainstorming, Python/Pandas workflow support, documentation, insight framing and presentation planning. Numerical outputs were checked against the analyzed results, including the incomplete December 2011 period.